

			0	
5	(a)	(i)	Correct directions with line of action of force vector passing through charges. (Arrow for negative charge pointing to the left and positive charge to the right)	A1
		(ii)	$F = qE$ $= (1.2 \times 10^{-15})(2.0 \times 10^3)$	
			$= 2.4 \times 10^{-12} \text{ N}$	A1
		(iii)	$\tau = Fd \sin \theta$	
			$= (2.4 \times 10^{-12})(2.5 \times 10^{-3}) \sin 35^\circ$	M1
			$= 3.4(4) \times 10^{-15} \text{ N m}$	A1
		(iv)	either rotates to align with the field or oscillates (about a position) clockwise / with the positive charge on the right of the centre of dust particle	B1

	(b)	(i)	electric force is negative of potential energy gradient	B1
		(ii)	The two charges have opposite signs since the force between the charges is attractive / negative.	B1
		(iii)	$\text{gradient} = \frac{-1.00 - (-3.00)}{(9.6 \times 10^{-10} - 4.4 \times 10^{-10})}$ $= 3.846 \times 10^9$	M1
			$F = 3.846 \times 10^9 \times 1.6 \times 10^{-19}$	C1
			$= 6.2 \times 10^{-10} \text{ N}$	A1
6	(c)		Let P be the contact point between the jockey and the wire XY	